Last Mile Connectivity

Version 1 — Dynamic Last Mile Assignment Model

1 Problem Description

Version 0 considered a static problem in which all passenger requests and vehicle positions were known beforehand and a single optimization was performed. Version 1 generalises this to a dynamic setting in which passenger requests arrive continuously over time and vehicles are in motion while decisions are being made.

The system operates in discrete time stamps of fixed length Δt (for example, 30 seconds). At every time stamp, the Version-0 static problem is solved on the passengers and vehicles currently in the system. Passengers who are not assigned in a given time stamp are carried over to the next time stamp, with their remaining waiting-time tolerance reduced by the elapsed interval. The Version-0 objective, constraints, decision variables, and greedy algorithm are reused without modification within each time stamp.

Scenario Description

- Passenger requests arrive continuously over time rather than all at once.
- Vehicles are in motion between time stamps; their positions evolve at a known speed.
- Each time stamp is treated as a Version-0 static sub-problem on the current pool.
- Unassigned passengers are not lost; they are carried over until either assigned or timed out.

2 Sets

The Version-0 sets are retained. A new set is added for time stamps:

- Passengers: $P = \{1, 2, \ldots, p\}$
- Vehicles: $V = \{1, 2, \ldots, v\}$
- Stops: $S = \{1, 2, \ldots, s\}$
- Soft routes: $R = \{1, 2, \ldots, r\}$
- Time stamps: $T = \{0, \Delta t, 2\Delta t, \ldots\}$

At each time stamp $t \in T$, the active passenger pool is denoted $P(t)$ and consists of two groups:

- $P_{\text{new}}(t)$: passengers whose requests arrived during $(t - \Delta t,\ t]$.
- $P_{\text{wait}}(t)$: unassigned passengers carried over from time stamp $t - \Delta t$.

so that $P(t) = P_{\text{new}}(t) \cup P_{\text{wait}}(t)$.

3 Parameters

Passenger Parameters (Version 0 retained, with one addition)

- o_p: origin location
- $S_p \subseteq S$: feasible pickup stops
- $l_p^{\max}$: maximum walking distance
- l_p^{ideal}: ideal walking distance
- d_p^{walk}: walking distance of passenger p
- T_p^{ideal}: ideal in-vehicle trip duration
- t_p^{req}: time at which passenger p submitted the ride request
- $W_p^{\max}$: maximum waiting-time tolerance (passenger cancels if exceeded)
- W_p: remaining waiting-time tolerance, initialised to $W_p^{\max}$ at request time and decremented by Δt at the end of every time stamp in which p is not assigned.

Vehicle Parameters (Version 0 retained, with the static origin generalised to a time-varying position)

- Cap_v: vehicle capacity
- R_v: base soft route
- $\Delta_v^{\max}$: maximum allowable detour
- C_v^{km}: operating cost per km
- $\text{pos}_v(t)$: current position of vehicle v at time stamp t
- ν_v: cruising speed of vehicle v (e.g., 30 km/h)

Weight Parameters

- γ: vehicle operating distance weight
- α: walking discomfort weight
- β: passenger trip duration weight
- μ: occupancy encouragement weight
- λ: rejection penalty weight

Stop Parameters

- loc_s: stop location

Derived Parameters

- $\delta_v(t)$: total passenger detour for vehicle v at time stamp t
- B_v: base route distance (origin $\rightarrow$ destination without detours) for vehicle v
- $n_v(t)$: number of passengers assigned to vehicle v at time stamp t

System Parameters

- Δt: time stamp interval (e.g., 30 seconds)

4 Decision Variables

All Version-0 decision variables are retained, now evaluated at each time stamp t.

Passenger–Vehicle Assignment

$$x_{pv}(t) = \begin{cases} 1, & \text{if passenger } p \text{ is assigned to vehicle } v \text{ at time stamp } t, \\ 0, & \text{otherwise.} \end{cases}$$

Rejection Penalty Indicator

$$r_p(t) = \begin{cases} 1, & \text{if passenger } p \text{ is rejected at time stamp } t, \\ 0, & \text{otherwise.} \end{cases}$$

Passenger Stop Assignment

$$x_{ps}^{\text{pickup}}(t) = \begin{cases} 1, & \text{if } p \text{ is picked up at stop } s \text{ at time stamp } t, \\ 0, & \text{otherwise,} \end{cases}$$

$$x_{ps}^{\text{drop}}(t) = \begin{cases} 1, & \text{if } p \text{ is dropped at stop } s \text{ at time stamp } t, \\ 0, & \text{otherwise.} \end{cases}$$

Vehicle Route Deviation $\delta_v(t)$ is the total additional distance travelled by vehicle v at time stamp t compared to its base route.

5 Constraints

All Version-0 hard and soft constraints are retained, now evaluated at each time stamp t. A single new constraint is introduced to handle waiting-time timeout.

Hard Constraints

If rejection is allowed (but penalised):

$$\sum_{v \in V} x_{pv}(t) \leq 1 \qquad \forall p \in P(t),$$

$$r_p(t) = 1 - \sum_{v \in V} x_{pv}(t) \qquad \forall p \in P(t)$$

Vehicle capacity:

$$\sum_{p \in P(t)} x_{pv}(t) \leq \mathrm{Cap}_v \qquad \forall v \in V$$

Passenger–stop linking:

$$\sum_{s \in S} x_{ps}^{\mathrm{pickup}}(t) = \sum_{v \in V} x_{pv}(t), \qquad \sum_{s \in S} x_{ps}^{\mathrm{drop}}(t) = \sum_{v \in V} x_{pv}(t) \qquad \forall p \in P(t)$$

Route deviation bound:

$$\delta_v(t) \leq \Delta_v^{\max} \qquad \forall v \in V$$

Waiting-Time Tolerance (new in Version 1)

$$\text{If } W_p \leq 0, \quad r_p(t) = 1$$

where

W_p is the remaining waiting-time tolerance of passenger p,

decremented by Δt at every time stamp in which p is not assigned.

Soft Constraints

Walking Discomfort Penalty

$$\mathrm{WalkingPenalty}_p = \alpha \, \max\Big(0, \; d_p^{\mathrm{walk}} - l_p^{\mathrm{ideal}}\Big)$$

where

α is the walking discomfort weight,

d_p^{walk} is the walking distance of passenger p,

l_p^{ideal} is the ideal walking distance for passenger p.

Trip Duration Penalty

$$\text{TripPenalty}_p = \beta \max\Big(0,\ T_p - T_p^{\text{ideal}}\Big)$$

where

β is the passenger trip duration weight,

T_p is the total in-vehicle trip duration of passenger p,

T_p^{ideal} is the ideal in-vehicle trip duration for passenger p.

Occupancy Penalty

$$\text{OccupancyReward}_v = -\mu\,(\text{Cap}_v - n_v(t))$$

where

μ is the occupancy encouragement weight,

Cap_v is the capacity of vehicle v,

$n_v(t)$ is the number of passengers assigned to vehicle v at time stamp t.

6 Objective Function

At each time stamp t, the objective function is identical in form to Version 0, evaluated on the active pool $P(t)$:

Let:

- B_v: base route distance of vehicle v (without detours)
- $\delta_v(t)$: total passenger detour for vehicle v at time stamp t
- $D_v(t) = B_v + \delta_v(t)$: total distance traveled by vehicle v at time stamp t
- $n_v(t)$: number of passengers assigned to vehicle v at time stamp t
- Cap_v: capacity of vehicle v
- C_v^{km}: operating cost per km of vehicle v
- d_p^{walk}: walking distance of passenger p
- T_p: total in-vehicle trip duration of passenger p
- l_p^{ideal}: ideal walking distance for passenger p
- T_p^{ideal}: ideal in-vehicle trip duration for passenger p
- γ: vehicle operating distance weight
- α: walking discomfort weight
- β: passenger trip duration weight

- μ: occupancy encouragement weight
- λ: rejection penalty weight
- $r_p(t)$: rejection indicator for passenger p at time stamp t

Then, for each time stamp t:

$$\min \ \gamma \sum_{v \in V} C_v^{\text{km}} \left(B_v + \delta_v(t)\right) + \alpha \sum_{p \in P(t)} \max\Big(0, d_p^{\text{walk}} - l_p^{\text{ideal}}\Big) + \beta \sum_{p \in P(t)} \max\Big(0, T_p - T_p^{\text{ideal}}\Big) - \mu \sum_{v \in V} \left(\text{Cap}_v - n_v(t)\right) + \lambda \sum_{p \in P(t)} r_p(t)$$

7 Dynamic Algorithm

Input:

- Stream of passenger requests $\{(p,\ t_p^{\text{req}})\}$ with their parameters.
- V: vehicle set with initial positions and base soft routes.
- S: stop set.
- Cap_v, $\Delta_v^{\max}$, ν_v, C_v^{km} for each vehicle.
- Δt: time stamp interval.
- $W_p^{\max}$ for each passenger.

Output:

- Time-indexed assignments $x_{pv}(t)$, $x_{ps}^{\text{pickup}}(t)$, $x_{ps}^{\text{drop}}(t)$.
- Updated vehicle routes and positions at every time stamp.
- Final list of served and rejected passengers.

Step 1: Update Passenger Pool

At each time stamp t, form the active pool:

- Add new passengers: any p with $t_p^{\text{req}} \in (t - \Delta t,\ t]$, initialised with $W_p = W_p^{\max}$.
- Carry over unassigned passengers from $P_{\text{wait}}(t)$, with $W_p \leftarrow W_p - \Delta t$.

The active pool is $P(t) = P_{\text{new}}(t) \cup P_{\text{wait}}(t)$.

Step 2: Retire Timed-Out Passengers

For each $p \in P(t)$, if $W_p \leq 0$, set $r_p(t) = 1$ and remove p from the active pool. Each such passenger contributes λ to the objective.

Step 3: Update Vehicle States

For each vehicle v, update $\text{pos}_v(t)$ based on its movement during the previous interval at speed ν_v. Vehicles physically advance along their routes between time stamps.

Step 4: Solve the Static Sub-Problem

Invoke the Version-0 greedy initialization algorithm on $P(t)$ and the current vehicle states. The Version-0 objective and all constraints apply unchanged. This produces:

- $x_{pv}(t)$: passenger–vehicle assignment.
- $x_{ps}^{\text{pickup}}(t),\ x_{ps}^{\text{drop}}(t)$: pickup and drop-off stops.
- Updated route and detour $\delta_v(t)$ for each vehicle.

Step 5: Carry Over Unassigned Passengers

Any passenger $p \in P(t)$ for whom $\sum_{v \in V} x_{pv}(t) = 0$ and $W_p > 0$ is added to $P_{\text{wait}}(t + \Delta t)$. Their W_p will be decremented in Step 1 of the next time stamp.

Step 6: Advance Time

Set $t \leftarrow t + \Delta t$ and return to Step 1.

Termination. In a real deployment, the loop runs indefinitely. For experimental evaluation, it terminates after a fixed simulation horizon T_{sim}.

Remark. Version 1 reuses the entire Version-0 model as an inner solver. The only genuinely new ingredients are: (i) a discrete time stamp set T, (ii) the carry-over rule with the decrementing W_p, and (iii) vehicles physically advancing at speed ν_v between time stamps. All other elements—objective, constraints, greedy algorithm, cost calculation—are identical to Version 0.

8 Cost Calculation

The system cost at each time stamp is computed using the Version-0 cost formula. The total cost over a simulation horizon is the sum over all time stamps:

$$C_{\text{total}} = \sum_{t \in T} \left[\gamma \sum_{v \in V} C_v^{\text{km}} \left(B_v + \delta_v(t)\right) + \alpha \sum_{p \in P(t)} \max\left(0, d_p^{\text{walk}} - l_p^{\text{ideal}}\right) + \beta \sum_{p \in P(t)} \max\left(0, T_p - T_p^{\text{ideal}}\right) - \mu \sum_{v \in V} \left(\text{Cap}_v - n_v(t)\right) + \lambda \sum_{p \in P(t)} r_p(t) \right]$$

Where:

- C_{total}: total system cost over the simulation horizon
- T: set of time stamps in the simulation horizon
- $P(t)$: active passenger pool at time stamp t

- γ: vehicle operating distance weight
- α: walking discomfort weight
- β: passenger trip duration weight
- μ: occupancy encouragement weight
- λ: rejection penalty weight
- B_v: base route distance of vehicle v
- $\delta_v(t)$: detour of vehicle v at time stamp t
- C_v^{km}: operating cost per km of vehicle v
- Cap_v: capacity of vehicle v
- $n_v(t)$: passengers assigned to vehicle v at time stamp t
- d_p^{walk}: walking distance of passenger p
- l_p^{ideal}: ideal walking distance for passenger p
- T_p: total in-vehicle trip duration of passenger p
- T_p^{ideal}: ideal in-vehicle trip duration of passenger p
- $r_p(t)$: rejection indicator at time stamp t

This value represents the cumulative objective minimized by the rolling Version-0 greedy across all time stamps.